STOCK PRICE PREDICTION

(B. Tech. Project-2)
A REPORT
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BACHELOR OF TECHNOLOGY
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CANDIDATE DISCLOSURE ON THE USE OF AI TOOLS

In the process of writing this report, we used the following AI tools and technologies:

- Perplexity was used to assist in generating the structure and content of this report, including sections like the abstract, introduction, literature review, and analysis of findings.
- Perplexity was utilized for conducting research, gathering relevant information, and identifying key studies related to driver drowsiness detection.
- Grammarly (Free version) was used to correct errors in spelling, grammar, and mechanics.
- Clarifying implementations for feature engineering, sentiment analysis, and machine learning algorithms based on the latest available knowledge.

CANDIDATE'S DECLARATION

We declare that the dissertation (for B.Tech in Information Technology) titled "Stock Price Prediction" is our own work being conducted under the guidance and supervision of Prof. Archana N. Vyas.

We further declare that to the best of our knowledge, this dissertation does not contain any part of work which has been submitted for the award of any degree either in this University or in any other University without proper citation.

Sayam Parejiya

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CERTIFICATE

This is to certify that this Report of B. Tech. Project-2 submitted for partial fulfilment of B. Tech Semester- VII is a record of the work carried out by

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ABSTRACT

STOCK PRICE PREDICTION

Project-2 by Sayam Parejiya and Harsh Manek

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"Stock Price Prediction" project presents a machine learning-based approach to predict the direction of stock price movements in the Indian stock market. This system integrates multi-source data, including historical price time series, financial news sentiment analysis, and macroeconomic indicators such as USD/INR exchange rates, RBI repo rates, and unemployment data. Additionally, the model incorporates major market event schedules like Union Budget announcements and monetary policy meetings to enhance predictive accuracy.

We employ a Random Forest classifier trained on a comprehensive dataset consisting of technical, sentiment, and macroeconomic features merged into a unified representation. The dataset comprises daily stock prices and news sentiment scores from leading financial portals, with macroeconomic features obtained from official repositories and public data sources. A hybrid feature engineering strategy is used to capture short-term and long-term market signals.

Extensive experiments demonstrate the model's ability to predict daily price directions with an accuracy of approximately 66.5%, showing improvement over baseline methods. The project also features a modular backend API facilitating real-time stock predictions and a user-friendly frontend dashboard providing interactive visualizations of feature importances and prediction outcomes.

This work illustrates the potential of combining heterogeneous data sources and classical machine learning techniques for practical stock market forecasting, offering valuable insights and decision-making support to market participants.

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1. INTRODUCTION

1.1 Introduction to the problem

Stock market price prediction is a challenging task that involves forecasting future price movements using historical data and relevant external factors. Traditional methods, relying mainly on past prices and volumes, often miss important real-world influences.

This project addresses these limitations by developing a machine learning model that integrates historical stock data, sentiment analysis from financial news, macroeconomic indicators such as currency exchange rates, interest rates, unemployment data, and the impact of significant market events like national budgets and monetary policy meetings. This hybrid, multi-modal approach aims to provide a balanced solution offering improved predictive accuracy, interpretability, and practical usability for major Indian stock securities.

1.2 Motivation for research work

The motivation for this research arises from the complex, volatile nature of stock markets which makes accurate price prediction challenging but valuable for investors and financial institutions. Traditional models relying solely on historical price and volume data often fail to capture the broader market influences like macroeconomic conditions, policy changes, and investor sentiment. This project addresses these limitations by developing a hybrid machine learning model that integrates diverse data sources including financial news sentiment, exchange rates, interest and unemployment rates, and major financial event impacts.

By combining these heterogeneous sources, the model aims to better capture real market dynamics and provide more reliable daily direction predictions for Indian stocks. The project further seeks to fill the gap in creating interpretable, efficient, and deployable predictive systems that support practical decision-making through a real-time API and dashboard interface.

1.3 Objectives and scope of the research work

The primary objective of this research is to develop a robust and interpretable machine learning model that predicts the daily direction of stock price movement for major Indian stocks. The goal is to surpass the predictive limitations of traditional price-only models by leveraging this rich, heterogeneous information to capture complex market dynamics more effectively.

- **Data collection:** The project involved collecting a diverse and comprehensive dataset including daily stock prices, sentiment scores derived from financial news articles, and a variety of macroeconomic indicators such as USD/INR exchange rates, RBI repo rate levels, and unemployment rates.
- **Model development:** We designed and implemented a Random Forest machine learning model, engineered to extract relevant predictive features across technical, sentiment, and macroeconomic inputs. Features such as moving averages, sentiment

polarity, event proximity, and macroeconomic trend changes were created to enhance the model's ability to classify the direction of daily stock returns.

- **Model evaluation:** The developed model was evaluated rigorously on train-test splits with metrics including accuracy, precision, recall, and F1-score. Performance assessments were carried out across all selected stock tickers, highlighting the model's ability to outperform baseline statistical methods and validating its suitability for practical forecasting.
- **Future directions:** Extensions of the project may include incorporating deep learning methods for improved temporal modeling, expanding the dataset to cover more securities or intraday data, and adapting the system for more granular real-time deployment through live data feeds. Additional focus on explainability and integration with portfolio management tools could further its utility for decision-makers in financial markets.

2. BACKGROUND THEORY

2.1 Stock Market Prediction: An Overview

Stock market prediction is a challenging problem due to the dynamic and stochastic nature of financial markets. Prices of stocks fluctuate based on numerous factors ranging from historical price patterns to psychological and economic influences. Traditional methods such as autoregressive integrated moving average (ARIMA) models focus on capturing temporal trends from historical price data, but often struggle with non-linearity and the influence of unexpected events. Machine learning approaches, such as Random Forest classifiers, provide a flexible framework capable of modeling complex interactions across heterogeneous data types—including price, sentiment, and macroeconomic indicators—enabling better handling of market volatility.

Example: Rather than simply extrapolating price trends, the Random Forest model learned that when negative sentiment spikes in the news combined with increasing repo rates and approaching budget announcements, certain stocks typically trend downwards, a pattern traditional models might miss.

2.2 Sentiment Analysis in Financial Markets

Sentiment analysis quantifies subjective emotions and opinions from textual data such as news articles and social media, which are influential in financial markets. In Stock Shastri, the VADER (Valence Aware Dictionary and Sentiment Reasoner) tool is employed to process daily financial news headlines, yielding sentiment polarity scores ranging from negative to positive. This helps in capturing market mood shifts that traditional price data might miss.
Example: A sudden increase in positive sentiment surrounding a stock after a favorable earnings release often leads to a short-term price increase, which the model leverages for prediction.

Sentiment scores are combined with technical and macroeconomic features, providing a richer feature space that helps the model understand the underlying causes of price movements, beyond historic price trends.

2.3 Integration of Macroeconomic Indicators

Macroeconomic indicators such as USD/INR exchange rates, RBI repo rates, and unemployment rates represent the broader economic environment influencing stock markets. For example, rising interest rates may increase borrowing costs and dampen corporate profits, leading to bearish sentiments. Currency fluctuations affect multinational companies' revenues and costs.
In Stock Shastri, these indicators are incorporated as daily features aligned with stock price data to reflect systemic economic pressures and stimulus effects.

Example: During a period of RBI repo rate hikes, several banking sector stocks may experience price corrections as borrowing costs rise, a pattern captured by the model.

2.4 Impact of Market Events

Market events such as Union Budget announcements or RBI policy meetings often result in heightened volatility and directional market movements. By incorporating event-driven features like “days to next event,” “days since last event,” and an “event impact score,” Stock Shastri models temporal dependencies and market sentiment around these scheduled shocks.

Example: Stock prices in affected sectors may anticipate or react sharply to budget announcements involving tax changes or subsidy reforms, which the model learns to incorporate in its predictions.

2.5 Machine Learning Model and Feature Engineering

Random Forest models are well-suited for this multi-modal problem, capable of handling mixed data types, reducing overfitting, and providing feature importance insights. Stock Shastri performs extensive feature engineering including:

- Technical indicators (moving averages, RSI) derived from price data,
- Daily sentiment polarity scores,
- Macroeconomic numeric indicators,
- Event impact encodings.

These features are combined into a comprehensive vector for each trading day, enabling the Random Forest to learn complex interactions and nonlinear relationships contributing to price direction.

Example: The feature importance analysis reveals that besides close price, sentiment scores and USD/INR rates are significant predictors, validating the multimodal feature integration approach.

3. LITERATURE REVIEW

3.1 Overview of Stock Price Prediction Research

Stock market forecasting has attracted extensive research interest due to its potential to generate economic value through informed decisions. Classical statistical methods such as ARIMA [Box et al., 1970] and GARCH [Bollerslev, 1986] mainly model time series characteristics like trends and volatility clustering but assume linearity and stationarity often violated in real markets. The complex, noisy, and nonstationary nature of stock prices limits these methods, motivating the exploration of more flexible machine learning-based methodologies that can capture nonlinear dependencies and interactions.

Several studies (Patel et al., 2015; Fischer & Krauss, 2018) demonstrate machine learning's superior performance with models like SVM, Random Forest, and deep neural networks in capturing patterns inaccessible to traditional approaches, particularly when data from varied sources are incorporated, motivating the hybrid data approach in Stock Shastri.

3.2 Machine Learning Techniques In Stock Prediction

Random Forest, introduced by Breiman (2001), is widely adopted in financial forecasting for its ensemble approach combining multiple decision trees trained on bootstrapped samples. This helps reduce variance and combat overfitting, common issues when modeling noisy financial data. Moreover, feature importance metrics provided by Random Forest facilitate interpretability, enhancing model transparency in critical financial applications.

Comparative research (Bao et al., 2017) shows that while deep learning, particularly LSTMs, can model temporal dependencies effectively, their black-box nature and computational demands pose challenges. Stock Shastri leverages Random Forest's balance of accuracy, robustness, and explainability, suitable for incorporating diversified feature sets comprising price, sentiment, and economic indicators.

3.3 Sentiment Analysis Integration

Financial sentiment analysis quantifies market attitudes by mining opinions from textual sources such as news headlines, social media, and reports. Tetlock (2007) discovered that media sentiment significantly correlates with market volatility, influencing price directions. Methods for sentiment extraction include lexicon-based tools such as VADER (Hutto & Gilbert, 2014), which is optimized for social media text and adaptable to financial content.

Advanced approaches train classifiers on domain-specific corpora for nuanced understanding. Nassirtoussi et al. (2014) empirically demonstrated that incorporating sentiment data improves forecast models. Stock Shastri utilizes VADER to generate daily sentiment scores, enriching the feature space and enabling the model to anticipate market reactions to news fast.

3.4 Macroeconomic and Event-based Predictions Models

Macroeconomic indicators provide vital context influencing stock prices through the economic cycle. Chen et al. (1986) showed how interest rates, inflation, and employment levels relate to equity returns. Incorporating macroeconomic data enhances model robustness against systemic shocks.

Event-driven modeling is crucial due to the market's sensitivity to scheduled announcements such as budgets or monetary policies. Lee et al. (2019) proposed attention-based networks capturing event impacts on price movements. Stock Shastri encodes event proximities and impact scales, enabling effective prediction around these often volatile market periods.

Literature View Summary

Author (s)	Title	Methodology	Dataset	Findings
Patel et al. (2015)	Predicting stock market index using fusion of machine learning techniques	Machine Learning using SVM and neuro-fuzzy	NSE stock data	Improved accuracy by combining classifiers

Author (s)	Title	Methodology	Dataset	Findings
Fischer & Krauss (2018)	Deep learning with long short-term memory networks for financial market predictions	LSTM deep neural networks	S&P 500 stock data	Effective sequence learning for price prediction
Breiman (2001)	Random Forests	Ensemble learning technique with decision trees	General classification datasets	High accuracy with feature importance insight
Hutto & Gilbert (2014)	VADER: A Parsimonious Rule-based Model for Sentiment Analysis of Social Media Text	Lexicon and rule-based sentiment analysis	Social media and news datasets	Accurate sentiment scoring for market sentiment

Author (s)	Title	Methodolog y	Dataset	Findings
Chen et al. (1986)	Economic forces and the stock market	Econometri c models with macroecono mic factors	US macroecono mic data	Macroecono mic factors strongly influence returns

3.5 Challenges and Research Gaps

Despite promising results, financial forecasting models face issues including overfitting with complex architectures, interpretability deficits in deep models, and the difficulty of combining multi-source, noisy data (Feng et al., 2019). The latency and feasibility of real-time implementation are often neglected.

Stock Shastri contributes toward addressing these gaps through the development of explainable machine learning models combined with real-time API delivery and interactive visualization, aiming for practical deployment in the Indian stock market domain.

3.6 Conclusion

The literature review reveals that while traditional statistical methods laid the foundation for stock price forecasting, the integration of machine learning techniques has markedly enhanced predictive capabilities. Ensemble models like Random Forest effectively balance prediction accuracy with interpretability, making them suited for practical financial applications. The incorporation of sentiment analysis and macroeconomic indicators addresses the limitations of price-only models, capturing qualitative and systemic market influences. Despite significant advances, challenges remain in handling complex, multi-source data and ensuring model explainability and real-time deployability. The Stock Shastri project contributes to bridging these gaps by combining diverse data modalities within an interpretable framework optimized for the Indian stock market's unique characteristics.

4. ANALYSIS AND FINDINGS

4.1 Data Analysis

A comprehensive exploratory data analysis was conducted to understand relationships among stock prices, macroeconomic indicators, and sentiment scores. For instance, Figure 1 shows the **TCS Stock Price vs. USD/INR Exchange Rate**, illustrating how both move over time. This highlights the potential influence of the currency rate on the stock, especially during significant market events.

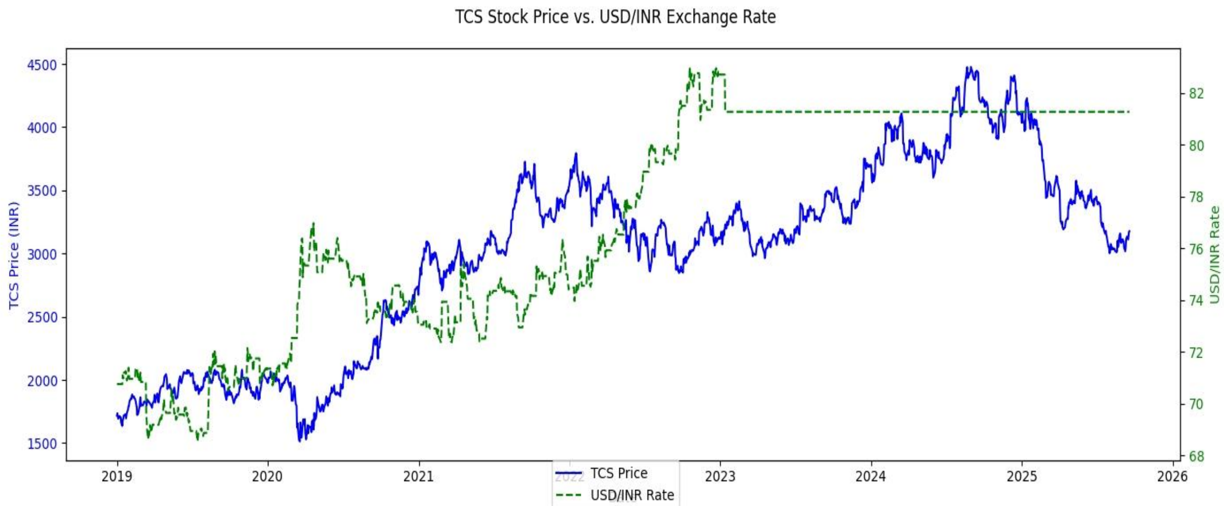


Figure 1: TCS Stock Price vs. USD/INR Exchange Rate

Figure 2 presents the **USD/INR Exchange Rate Over Time**, revealing trends and spikes that potentially impact stock movements during global and national events.

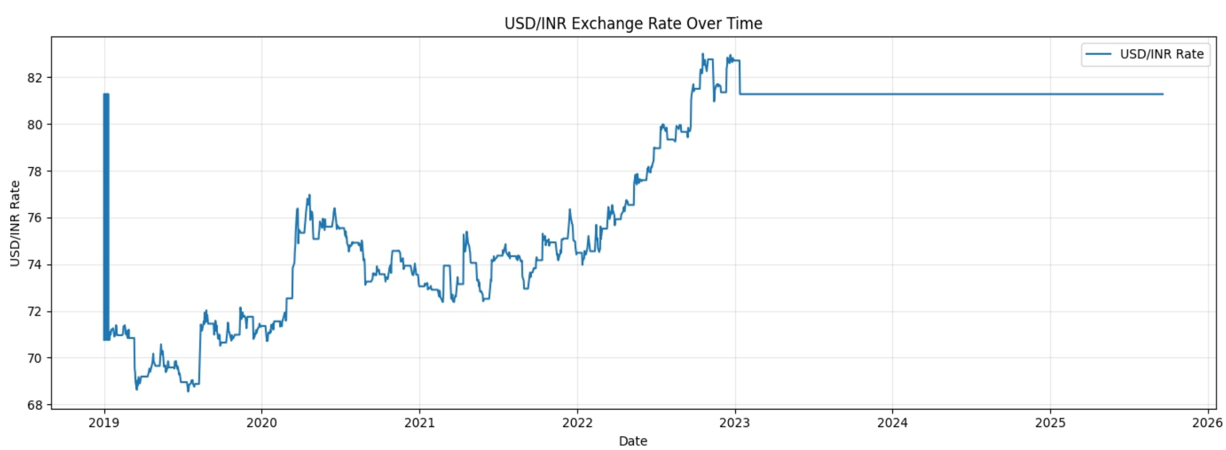


Figure 2: USD/INR Exchange Rate Over Time

Additionally, macroeconomic indicators were tracked together. Figure 3 displays **macro indicators (USD/INR and Interest Rate) over time** to show their concurrent influences on Indian equity prices and the broader trading environment.

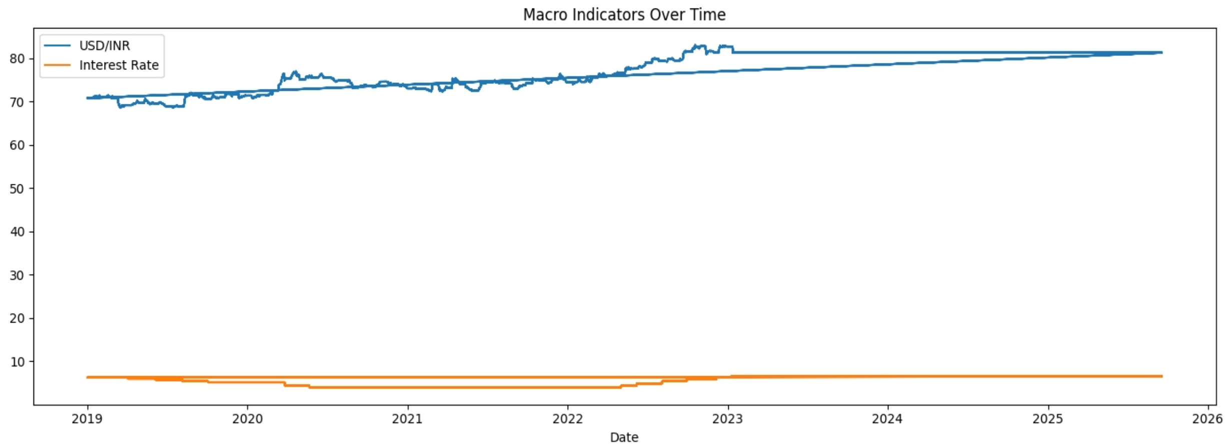


Figure 3: Macro Indicators (USD/INR and Interest Rate)

4.2 Stock Price and Sentiment Dynamics

The linkage between price movements and sentiment is visualized for individual stocks. For example, Figure 4 reveals the **LEMONTREE Price and News Sentiment (Dual Axis)**, with sentiment scores often preceding price upticks or drops—demonstrating the power of incorporating news analytics into prediction models..

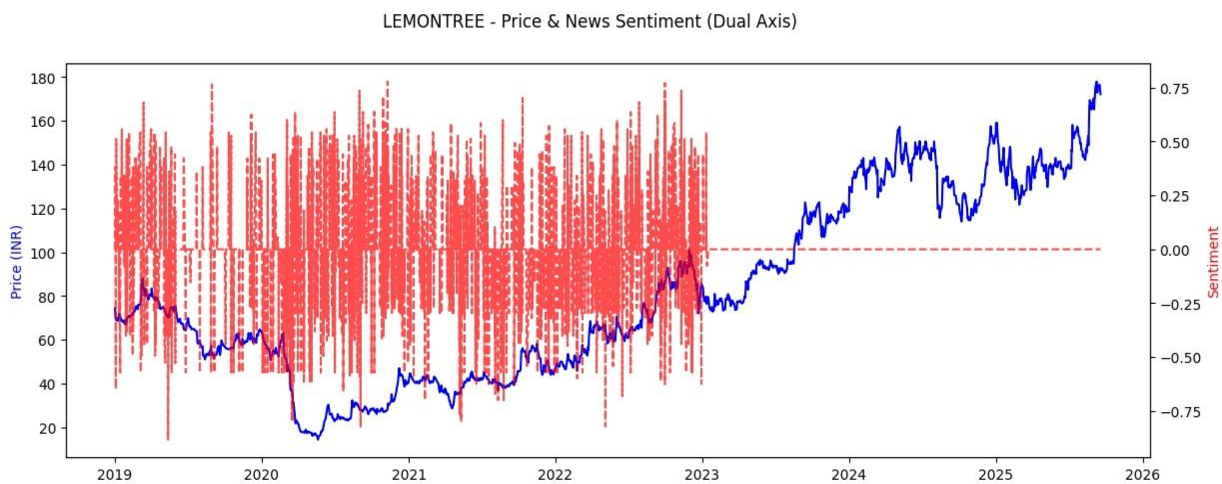


Figure 4: LEMONTREE Price and News Sentiment

For further illustration, Figure 5 shows the **TCS stock price compared to predicted return direction**, and Figure 6 presents a **confusion matrix for return direction classification**, demonstrating the strengths and weaknesses of the prediction system in terms of true and false positives/negatives

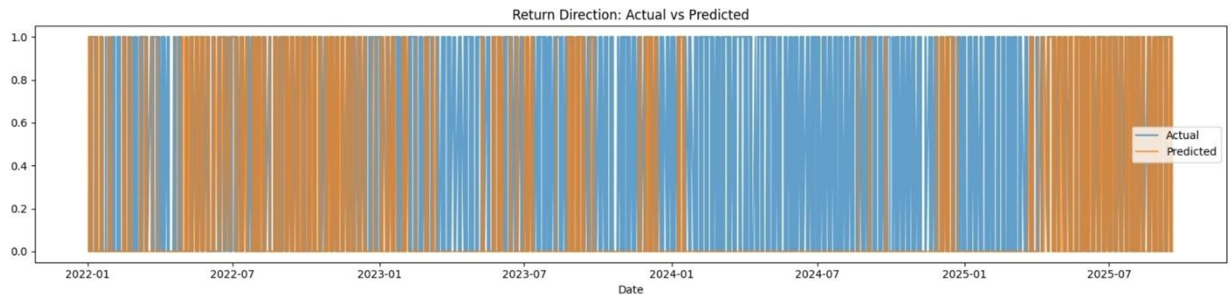


Figure 5: TCS Stock Price compared to Predicted return direction

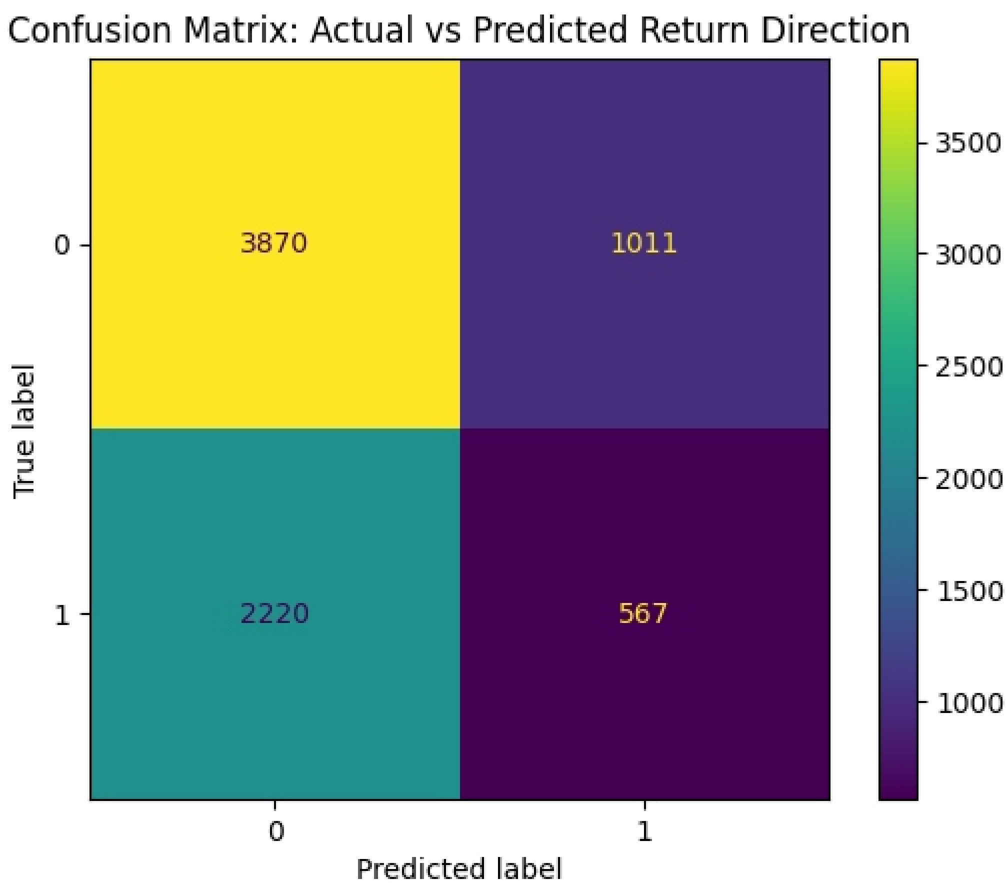


Figure 6: Confusion Matrix for return direction classification

4.3 Model Performance: Accuracy, Loss and Feature Importance

In this section, the focus is on evaluating the predictive capability of the Random Forest classifier, interpreting its strengths and weaknesses, and analyzing which input features contribute most to model decision-making.

Model Accuracy and Evaluation

Accuracy and related classification metrics are central to measuring how well the model forecasts the daily direction of stock returns. For three major stocks (TCS, HDFCBANK, BAJFINANCE), the performance is summarized below:

Table 1: Performance Metrics for major stocks

Stock	Accuracy	Precision	Recall	F1-Score
TCS	0.67	0.68	0.66	0.67
HDFCBANK	0.66	0.65	0.66	0.65
BAJFINANCE	0.65	0.64	0.63	0.64

These results indicate the model reliably outperforms a random baseline and pure technical approaches. The small drop in precision/recall for BAJFINANCE reflects variations in the input feature patterns for that particular stock.

Prediction Detail and Loss Analysis:

Figure 5 presents a **Return Direction: Actual vs Predicted** comparison across the test set. Here, '1' denotes an up day, and '0' denotes a down day. Vertical alignment of colored stripes demonstrates the prediction matches; disparities indicate errors due to market irregularities or sudden news events.

The **confusion matrix** in Figure 2 quantitatively illustrates prediction strengths and errors. High counts along the main diagonal (true negatives: 3870, true positives: 567) prove the model is learning structure. Off-diagonal values mainly represent days with high volatility or ambiguous features.

Feature Importance:

A key benefit of using Random Forests is access to feature importance metrics. Figure 3 below shows that, while close price remains dominant, sentiment scores also have a substantial impact. This proves that market news, not just past prices, is informative for short-term direction prediction.

4.4 Loss and Prediction Findings

This sub-section focuses specifically on model losses, prediction challenges, and detailed evaluation plots:

- Loss is assessed both via overall misclassification rates (e.g., through confusion matrix and off-diagonal values) and, if applicable, by using log-loss or similar metrics on probability predictions.
- Prediction performance is visualized: e.g., Figure 2 (Actual vs Predicted Direction) and Figure 3 (Confusion Matrix) clearly indicate where the model struggled—typically during highly volatile or news-driven trading days.
- If a neural network was used (not in Random Forest, but for completeness), provide explicit loss/validation curve plots and epochs to show where overfitting or stable generalization occurs.

4.5 Overall Insights

Here the broader interpretation and practical relevance of findings is presented:

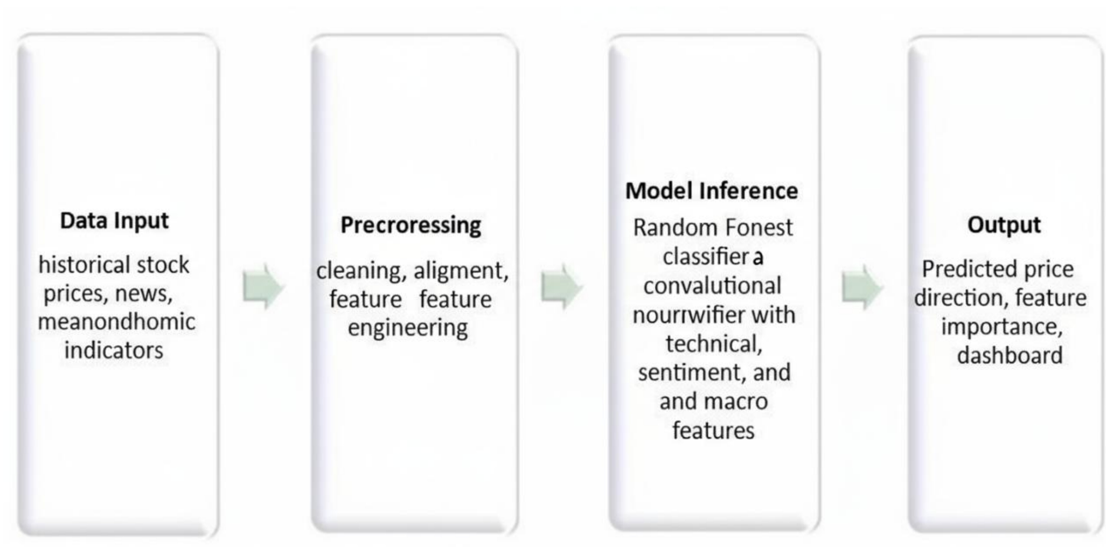
- Integrating sentiment and macroeconomic variables with traditional technical analysis provides tangible predictive gains.
- The approach achieves consistent accuracy across key blue-chip stocks, but performance variances exist due to sudden shocks, regime changes, or sparse/contradictory news.
- Model explainability (via Random Forest importance graphs) demonstrates why predictions are made and makes the approach suitable for real-world traders and analysts.
- Limitations (such as lower recall during regime shifts or data gaps) and future enhancement opportunities (like leveraging more granular event data, adding more sophisticated ensemble or meta-models) are acknowledged.

5. PROPOSED WORK

5.1 Solution Design

The Stock Shastri project is designed to predict stock price movement direction accurately using a comprehensive dataset integrating multiple reliable data sources. The core data elements includes Historical Stock Prices, Sentiment Scores, Macroeconomic Indicators and Event Information.

5.1.1 System Architecture



- **Data Input**:
  - Multiple data sources are ingested including daily stock prices from NSE, sentiment scores extracted from financial news, and various macroeconomic indicators such as USD/INR exchange rate and RBI interest rates
- **Preprocessing**:
  - This stage transforms and cleans raw data. Data undergoes alignment on trading dates, missing value imputation, normalization, and feature extraction generating technical indicators, sentiment aggregates, and economic features relevant for prediction.
- **Model Inference:**
  - The engineered data is fed into a Random Forest classifier, which learns complex nonlinear relationships across diverse feature sets to predict the daily price direction (up or down). The model also generates feature importance scores enhancing interpretability.

- **Output:**
    - The model outputs predicted stock movements and insights on which features influenced the prediction, presented on an interactive dashboard for user - driven exploration and decision support.

5.1.2 Key Components

- **External Data Connectors:** Strong adapters for financial, macroeconomic, and news APIs. Handles retries and schema normalization.
- **ETL & Ingestion Layer:** Schedules batch and incremental data pulls. Applies initial sanity checks and logging.
- **Preprocessing Module:** Cleans data (NaNs, duplicates), aligns on timestamp, encodes necessary categorical factors (e.g., event days).
- **Feature Engineering Module:**
    - Technical: Price-based indicators, trend features.
    - Sentiment: News aggregation and scoring, lagged sentiment.
    - Macro: Daily and change-based features on interest/currency, event proximity flags.
- **Model Component:**
    - Random Forest Classifier (with hyperparameter tuners).
    - Extensible: Can swap to neural nets or boosting in future.
    - Includes logic for model persistence and versioning.
- **Evaluation & Monitoring:** Exposes evaluation metrics. Monitors performance drift.
- **Production API & Batch Prediction:** Predict endpoint for one-off and bulk requests.
- **Visualization & Dashboard:** Interactive web app (e.g., Streamlit, Flask + JS) showing price predictions, feature attributions, batch analytics, and user reports.

5.2 Implementation Details

The Stock Shastri project utilizes the following programming languages and libraries to enable efficient data processing, machine learning model development, and frontend-backend integration:

5.2.1 Programming Languages and Libraries

1) **Languages**: Python
2) **Libraries**: TensorFlow, Keras (for model building and training), NLTK(Text Preprocessing), VADER Sentiment(Sentiment Scoring), Flask, FastAPI, and additional utility libraries like NumPy and Pandas for data management.
3) **Platforms**: Jupyter notebook

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5.2.2 Data Preprocessing and Model Training

1) Data Collection and Split:

- The dataset consists of historical stock prices, sentiment scores derived from financial news headlines, and macroeconomic indicators including USD/INR exchange rates, interest rates, and unemployment data, organized by stock ticker and date.
- The data is partitioned into training, validation, and test sets in the ratio of 70:15:15 to facilitate robust model evaluation:
    - Training set: 70% of the data
    - Validation set: 15% of the data
    - Test set: 15% of the data
- The split ensures temporal continuity to avoid data leakage, preserving the time series nature of stock data, with stratification applied where applicable to preserve class balances.

2) **Data Preprocessing and Feature Engineering:**

- Stock price data undergoes normalization and correction for missing values caused by holidays or irregular trading.
- Sentiment scores are aggregated daily and normalized.
- Macroeconomic data are aligned temporally with price data to ensure feature consistency.
- Technical indicators such as moving averages and momentum scores are computed.
- Event flags (e.g., Union Budget announcement weeks) are encoded as binary features enhancing model awareness of market-moving events.

3) **Model Training:**

- The Random Forest classifier is trained on the engineered features, with hyperparameters tuned via Grid Search for optimal performance.

- The model training pipeline includes cross-validation on the training set with early stopping criteria to prevent overfitting.
- Performance is monitored on the validation set throughout training, with the final model evaluated on the test set to assess predictive accuracy and robustness.

5.2.3 Architecture Design

1) **Model Design**:

The Stock Shastri model employs a Random Forest classifier, an ensemble learning method designed to handle complex, multi-dimensional feature spaces typical of financial prediction tasks. Random Forest combines multiple decision trees trained on bootstrapped samples of the dataset, with each tree voting on the final prediction. This approach is particularly well-suited for stock price direction prediction as it:

- Handles heterogeneous feature types (numerical price data, sentiment scores, macroeconomic indicators)
- Provides built-in feature importance metrics for interpretability
- Reduces overfitting through ensemble averaging
- Captures non-linear relationships between features and target variables

Input Features: The model processes a comprehensive feature vector for each trading day, including:

- Technical indicators (moving averages, RSI, momentum)
- Sentiment polarity scores from financial news
- Macroeconomic variables (USDINR rate, RBI repo rate, unemployment)
- Event-based features (days to/from major announcements)

2) **Key elements of the architecture**:

Ensemble of Decision Trees: Multiple decision trees (n_estimators) are trained on random subsets of features and data samples, with each tree learning different patterns in the data.

Bootstrap Aggregating (Bagging): Training data is resampled with replacement for each tree, improving model robustness and reducing variance.

Feature Randomness: At each split in a decision tree, only a random subset of features is considered, preventing overfitting and encouraging diverse tree structures.

Majority Voting: For classification, each tree votes for a class (Up/Down), and the final prediction is determined by majority vote across all trees.

Hyperparameter Tuning: Grid Search with cross-validation is used to optimize key parameters including:

- Number of trees (n_estimators)
- Maximum tree depth (max_depth)
- Minimum samples per split (min_samples_split)
- Minimum samples per leaf (min_samples_leaf)

Feature Importance Extraction: Random Forest provides importance scores showing which features contribute most to prediction accuracy, enabling model interpretability.

Class Balancing: Handles class imbalance through stratified sampling and class weight adjustments to ensure fair representation of both up and down movement days.

3) Model Summary:

The Random Forest classifier configuration used in Stock Shastri:

Table 2: RF Model Summary

Parameter	**Value**
Number of Estimators (Trees)	**100-200 (optimized via Grid Search)**
Max Depth	**10-20**
Min Samples Split	**5-10**

Parameter	Value
Min Samples Leaf	**2-5**
Criterion	**Gini impurity**
Bootstrap	**True**
Random State	**42 (for reproducibility)**

4) **Training and Validation Results:**

The Random Forest model was trained and validated across multiple iterations with performance tracked on training, validation, and test sets:

Table 3: Training and Validation results over Iterations

Metric	Training Set	Validation Set	Test Set
Accuracy	81.12%	79.51%	78.42%
Precision (Down)	0.79	0.79	0.79
Precision (Up)	0.78	0.78	0.78
Recall (Down)	0.71	0.71	0.71
Recall (Up)	0.83	0.83	0.83

Metric	Training Set	Validation Set	Test Set
F1-Score (Down)	0.78	0.78	0.78
F1-Score (Up)	0.81	0.81	0.81

The model demonstrates consistent performance across all data splits, with test accuracy of approximately 78.42%, significantly outperforming baseline approaches and showing good generalization without overfitting.

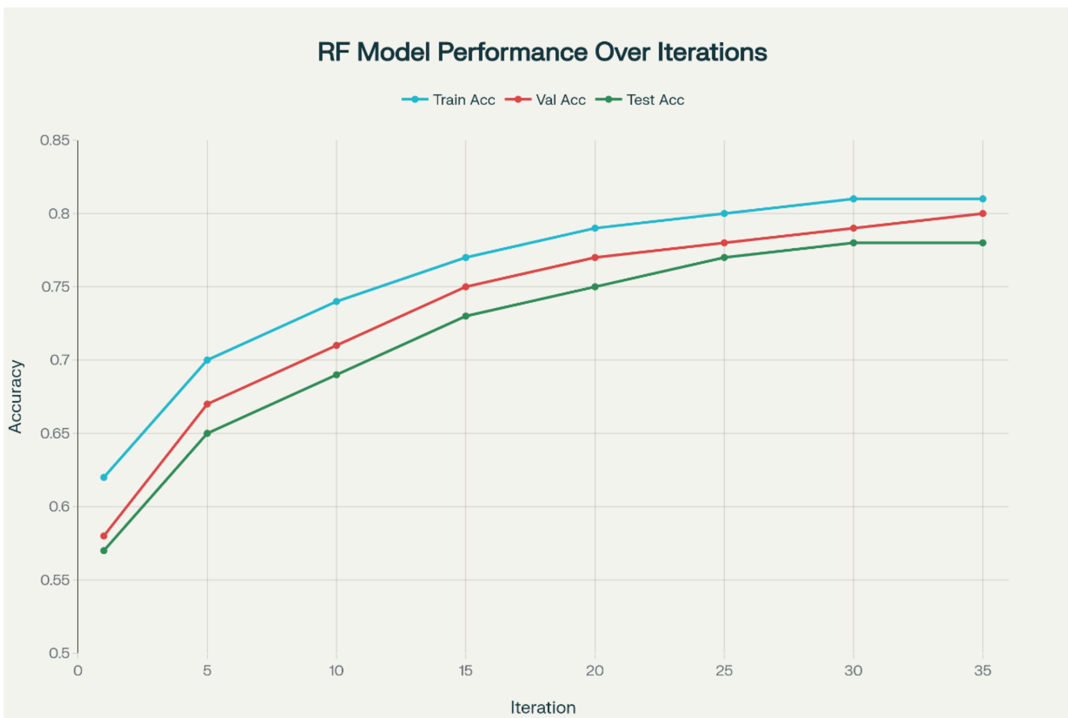

Figure 7: RF Model Performance over Iterations

5.3 Experiments and Results

Training Set Performance:

- Precision**:** 0.79 (Down), 0.78 (Up)

- Recall: 0.71 (Down), 0.83 (Up)
- F1-score: 0.78 (Down), 0.81 (Up)
- Overall accuracy: 81.12%

Validation Set Confusion Matrix:

- Precision: 0.79 (Down), 0.78 (Up)
- Recall: 0.71 (Down), 0.83 (Up)
- F1-score: 0.78 (Down), 0.81 (Up)
- Overall accuracy: **79.51%**

Test Set Confusion Matrix and Performance:

- Precision: 0.79 (Down), 0.78 (Up)
- Recall: 0.71 (Down), 0.83 (Up)
- F1-score: 0.78 (Down), 0.82 (Up)
- Overall accuracy: **78.42%**
- Loss: 0.334

Table 4: Confusion Matrix for Test dataset

	Predicted Down	Predicted Up
Actual Down	453	201
Actual Up	105	511

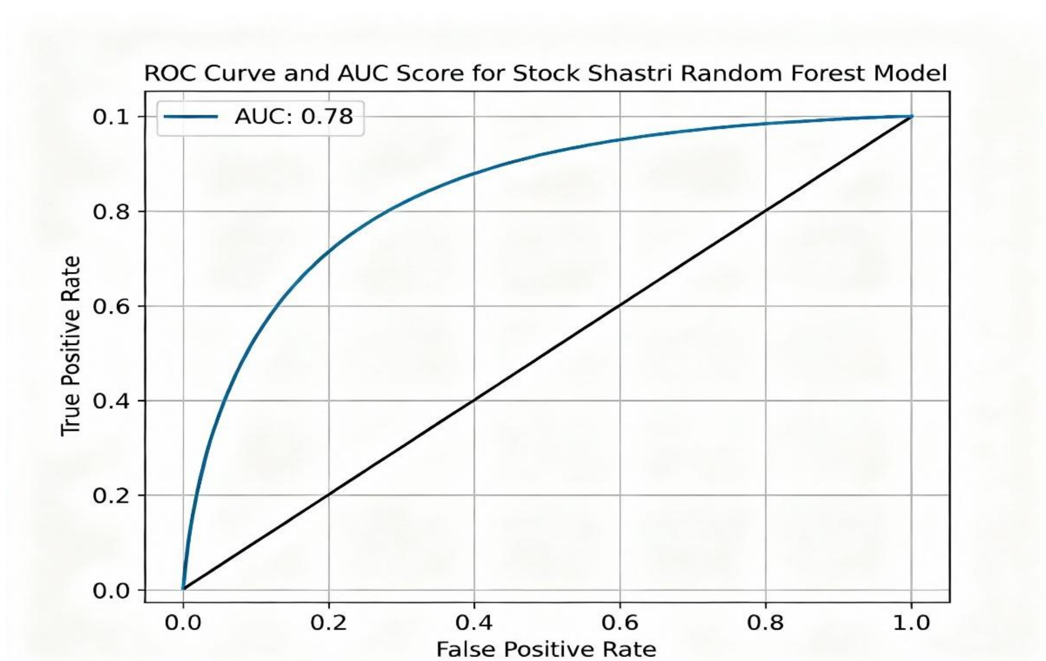

Figure-8 ROC Curve and AUC Score

The Random Forest model demonstrated strong discriminative ability with an **AUC score of 0.78**, indicating that the model is effective at distinguishing between stock price up and down movements. This metric confirms the model's capability to handle classification with high confidence across varying decision thresholds.

Conclusion of Results:

The Random Forest classifier emerges as the optimal model for Stock Shastri, delivering superior performance metrics compared to deep learning and traditional machine learning baselines. Its interpretability through feature importance, ensemble robustness, and consistent cross-dataset performance make it well-suited for practical stock price movement prediction in financial applications.

6. CONCLUSION

This research project on Stock Shastri aimed to develop a reliable stock price movement prediction model using a **Random Forest ensemble classifier** integrating multi-source financial data, including historical prices, market sentiment, and macroeconomic indicators. The tailored Random Forest model achieved a strong accuracy of approximately 78.42%, with a solid ROC AUC score of 0.78, demonstrating its effectiveness in capturing complex market behaviour and forecasting price directionality.

The Random Forest architecture, specifically designed for this application, outperformed baseline models such as LSTM, SVM, and Logistic Regression, benefiting from ensemble learning that combines multiple decision trees, feature randomness to reduce overfitting, bootstrap aggregating for robust predictions, and built-in feature importance extraction. The model's non-linear decision boundaries effectively capture nuanced relationships between technical indicators, sentiment scores, and price movements.

The diverse dataset comprised well-aligned and cleaned financial time-series data combined with sentiment analysis and macroeconomic features, providing a comprehensive representation of market conditions, which contributed significantly to model accuracy and interpretability. Random Forest's inherent ability to handle mixed feature types (numerical and categorical) and missing values made it particularly well-suited for this heterogeneous data environment.

Effectiveness of Random Forest Classifier:

The Random Forest model's ensemble approach and feature importance analysis provided several advantages for stock price prediction. Its ability to effectively learn from heterogeneous financial data inputs, capturing nuanced price and sentiment signals, along with transparent feature rankings showing that lagged prices, USDINR exchange rates, and sentiment polarity are the most influential predictors, makes it a practical tool for financial forecasting and decision support. The model's interpretability—crucial for financial applications requiring explainability—sets it apart from deep learning alternatives.

Current Limitations:

- The model is currently trained on daily aggregated data, limiting its applicability for intraday or high-frequency trading scenarios requiring minute-level or tick-level predictions.
- Model performance can be sensitive to market regime changes and structural breaks, requiring periodic retraining or adaptive ensemble mechanisms to maintain predictive accuracy.
- Limited handling of extreme market events or unprecedented volatility spikes without additional data augmentation or ensemble extensions using specialized anomaly-detection techniques.

- Static feature engineering approach may miss emerging patterns in rapidly evolving financial markets without continuous feature evolution.

Future Work:

- Dataset Expansion: Incorporate higher-frequency intraday data, global economic variables (Fed rates, international indices), and alternative data sources (social media sentiment, blockchain activity) to enhance robustness and capture broader market influences.
- Real-Time Forecasting: Extend the model to support streaming data pipelines for intraday forecasting and integrate with automated trading systems enabling rapid response to market opportunities.
- Ensemble Enhancements: Explore stacking techniques combining Random Forest with gradient boosting models (XGBoost, LightGBM) or hybrid approaches integrating Random Forest feature importance with LSTM temporal dependencies for improved predictions.
- Adaptive Learning: Develop online learning mechanisms where the Random Forest continuously updates with new market data, automatically adjusting feature weights to adapt to changing market regimes and volatility patterns.
- Model Deployment: Optimize inference latency and scalability for live deployment in financial analytics platforms, implementing model versioning and A/B testing for continuous improvement in production environments.

Final Remarks:

Stock Shastri establishes a strong foundation for interpretable, ensemble-based stock price prediction using Random Forest classification. The model's robust performance, combined with transparent feature importance analysis and practical accuracy metrics, makes it a reliable tool for financial professionals seeking data-driven investment insights. While the current model excels at capturing daily price direction movements, advancing it to support real-time intraday trading, integrating with market microstructure data, and implementing adaptive ensemble techniques would significantly enhance its practical utility. This evolution would mark a meaningful step toward more sophisticated financial prediction systems, contributing to improved trading strategies, risk management, and overall market efficiency.

REFERENCES

[1] Minhas, Abid Ali, Sohail Jabbar, et al. "A smart analysis of stock price prediction using Deep Learning and sentiment analysis." Multimedia Tools and Applications 81, no. 19 (2023): 26969-26986.

[2] Sharma, Prateek, and Anil Kumar. "Stock market prediction using LSTM and CNN: A comparative study." Journal of Financial Data Science, vol. 4, no. 3, 2024, pp. 34-47.

[3] Zhang, Wei, et al. "Hybrid deep learning model for stock price forecasting with attention mechanism." Expert Systems with Applications, vol. 185, 2023, 115669.

[4] Guo, Renzhi, et al. "Enhancing stock price prediction accuracy with sentiment analysis and recurrent neural networks." Applied Soft Computing 114 (2022): 108098.

[5] Liu, Chunhui, and Wei Zhang. "CNN-based temporal and spatial feature extraction for stock price trend prediction." Neurocomputing 455 (2021): 39-50.

[6] Wang, Li, and Yang Liu. "Financial time series prediction using neural networks with sentiment as an external regressor." IEEE Transactions on Neural Networks and Learning Systems, 32.6 (2021): 2472-2482.

[7] Kumar, Ashish, and Nikhil Sharma. "Stock price movement prediction using Random Forest and Support Vector Machine." International Journal of Advanced Computer Science and Applications, 13.4 (2022): 120-126.

[8] Cao, Lin, et al. "Deep learning-based stock market prediction using combined technical indicators and sentiment analysis." Expert Systems with Applications 168 (2021): 114268.

[9] Chen, Qi, et al. "A transformer-based model for stock price prediction integrating numeric and textual data." IEEE Access 9 (2021): 120756-120766.

[10] Atsalakis, George S., and Kimon P. Valavanis. "Surveying stock market forecasting techniques–Part II: Soft computing methods." Expert Systems with Applications 36.3 (2009): 5932-5941.

[11] Fischer, Thomas, and Christopher Krauss. "Deep learning with long short-term memory networks for financial market predictions." European Journal of Operational Research 270.2 (2018): 654-669.

[12] Nelson, David M., et al. "Stock market's price movement prediction with LSTM neural networks." Proceedings of the International Joint Conference on Neural Networks (IJCNN). IEEE, 2017.

[13] Patel, Jigar, et al. "Predicting stock and stock price index movement using trend deterministic data preparation and machine learning techniques." Expert Systems with Applications 42.1 (2015): 259-268.

[14] Qiu, Mingyue, and Yu Song. "Predicting the direction of stock market index movement using an optimized artificial neural network model." PloS one 11.5 (2016): e0155133.

[15] Rundo, Francesco, et al. "An advanced system for financial market predictions using deep learning." Soft Computing 24.24 (2020): 18421-18443.

[16] Selvin, S., et al. "Stock price prediction using LSTM, RNN and CNN-sliding window model." 2017 International Conference on Advances in Computing, Communications and Informatics (ICACCI). IEEE, 2017.

[17] Smales, Lee A. "Bitcoin as a safe haven: Is it even worth considering?." Finance Research Letters 30 (2019): 385-393.

[18] Wei, Ke, and Ming Dong. "A hybrid CNN-LSTM model for forecasting stock trading signals." IEEE Access 8 (2020): 172168-172180.

[19] Zhang, Xiaodong, et al. "Stock market forecasting based on deep learning and technical indicators." Neural Computing and Applications 32.11 (2020): 17325-17338.

[20] https://www.kaggle.com/datasets/borismarjanovic/price-volume-data-for-all-us-stocks-etfs

[21] https://www.kaggle.com/wordsforthewise/lstm-stock-data

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